



INTRODUCTION

Morton's Horticultural Products (MHP) offers its employees the ability to buy stock options as a means of additional compensation. Joseph, a prospective employee, has been offered 20,000 options with a strike price of \$30 that will vest in four years. While stock options can be a valuable form of compensation for employees, Joseph knows that they do not come without risk. To ensure that he is making a worthwhile decision, Joseph wants to conduct a financial analysis regarding the MHP stock. The issue is whether Joseph's valuation method correctly reflects the economic value of the options and whether his assumptions are consistent with no-arbitrage pricing principles. Since the stock valuation at the end of four years is ultimately unknown, this analysis will be completed using expected future value and risk-adjusted valuation techniques. Joseph has tasked our team with completing basic and advanced financial analyses of MHP's stock to help him make a sound financial decision.

1. PROBLEM STATEMENT AND ASSUMPTIONS

The purpose of this report is to evaluate the expected value of Morton's Horticultural Products stock at a series of key points in the future and to determine the fair value of the associated employee stock options. This report will determine the value of one share of MHP stock at the end of one year and then at the end of four years under Joseph's assumptions. This analysis will help Joseph determine whether his investment in the stock options will be profitable, whether Joseph's expected value logic is correct, and it will help Joseph understand the risk involved in this decision. By using financial tools such as Time Value of Money, Expected Future Value, Net Present Value, and the Binomial Asset Pricing Model, we will determine the fair value of both the stock and stock option discounted to 2013, compare Joseph's prediction with the no-arbitrage valuation implied by the Binomial Asset Pricing Model, provide an analysis of Mark's investment, and provide an analysis of the relationship between the stock option valuation and the probability of MHP stock increasing or decreasing in value.

2. SUMMARY OF RESULTS

This analysis evaluated whether Joseph's valuation approach is consistent with no-arbitrage pricing principles. The first was Joseph's expected value method based on equal probabilities for the stock to move up or down. The second was the binomial asset pricing model, which uses a risk-neutral probability. Both methods assume MHP stock moves 30% up or 10% down over the risk-free rate of 5% each year.

Joseph's method assigns a 50% probability to the up and down movements. Using this assumption, the expected Year 4 stock price is \$53.40, which discounts to \$43.92 in present value. The expected option payoff at Year 4 is \$23.77 per share. Discounting this expected payoff at the 5% risk-free rate over four years yields a present value of \$19.56 per option. Applied to a one-year option, this method produces a value of \$5.21.

The binomial model assumes the same stock movements up and down but calculates a risk-neutral probability of $p=0.25$ from the formula $p=(1+r-d)/(u-d)$. This probability makes the expected stock return equal the risk-free rate. This adjustment eliminates arbitrage opportunities and ensures that the option price is consistent with market equilibrium. Using $p=0.25$, the expected Year 4 option payoff is \$8.39,



which discounts to \$6.90 per share and \$138,000 for all 20,000 shares. The one-year option under this model is worth \$2.61.

Joseph's valuation exceeds the binomial price by \$12.66 per share for the four-year option, overvaluing the total 20,000 shares by approximately \$253,000. For the one-year case, Joseph's \$5.21 materially exceeds the no-arbitrage value of \$2.61.

Mark's arbitrage trade shows why the binomial price is correct. Mark buys 9,125 shares using \$27,375 of his own money and borrowing \$246,375 at 5%. He then sells 10,500 one-year options to Joseph at Joseph's price of \$5.21. This position is constructed so that Mark's outcomes are the same whether the stock goes up or down. If the stock goes to \$40.95, his shares gain value but he pays out on the options. If it falls to \$28.35, his shares lose value but the options expire with no value. After accounting for loan repayment and premium reinvestment Mark will profit \$27,330 in both scenarios (premium + borrowing – share purchase). Because the payoff is identical in both states, Mark earns a return exceeding the risk-free rate without bearing state-dependent risk.

If the option were priced at its fair value of \$2.61, Mark wouldn't make any money from the trade — not when he sets it up and not when it finishes. But if the price is anything higher than \$2.61, Mark can lock in a guaranteed profit no matter what the stock does.

The issue with Joseph's method is that he discounts risky payoffs at the risk-free rate while using subjective probabilities. The stock price of \$30 already incorporates the investors' beliefs about the future of the stock. Option prices must be derived from the stock price through no-arbitrage conditions. Joseph's 50/50 assumption does not affect the fair option price because those beliefs are already reflected in the \$30 stock price.

3. ANALYSIS

While both approaches generate the same discounted expected stock value, they produce materially different option valuations due to the treatment of risk and probability weighting. These approaches included an expected value approach using equal up/down probabilities (Joseph method) and a binomial asset pricing approach using risk-neutral probabilities and hedge-consistent weighting.

Four-Period Stock Price and Option Payoff Tree					
Year 0	Year 1	Year 2	Year 3	Year 4	Year 4 Pay Off
				\$ 104.15	\$ 74.15
			\$ 76.30		
		\$ 55.90		\$ 72.10	\$ 42.10
	\$ 40.95		\$ 52.82		
\$ 30.00		\$ 38.70		\$ 49.92	\$ 19.92
	\$ 28.35		\$ 36.57		
		\$ 26.79		\$ 34.56	\$ 4.56



			\$ 25.32		
				\$ 23.92	-

Figure 1 Stock Price and Option Payoff Tree

Under both approaches, the expected Year 4 stock value is the same when discounted to present value, at approximately \$43.92 per share (Figure 1 and 2). This supports the concept that stock assumptions are consistent across models. The difference appears in the option valuation, not the stock forecast.

$$E[S_4] = S_0 * \left(\frac{u+d}{2}\right)^4 = (30) * 1.155^4 = \$53.38 \mid PV(E[S_4]) = \frac{53.38}{1.05^4} = \$43.92$$

Using Joseph's expected value method with equal 50/50 annual probabilities, the expected Year-4 option payoff is \$23.77, and its present value is \$19.55 after discounting at 5% for four years (Figure 2). Applied to 20,000 options, this implies a total present value of about \$391,000. (Figure 2).

Metric	Joseph Method	Binomial Model
Present value of stock	\$ 43.92	\$ 43.92
Option value per share	\$ 19.55	\$ 6.90
Options granted	20,000.00	20,000.00
Total option value	\$ 391,086.00	\$ 138,007.00

Figure 2 Option Valuation Summary Table

Using the binomial asset pricing model, the same stock lattice tree was evaluated. The probability-weighted expected payoff falls to \$6.90. This applied to 20,000 shares equates to \$138,007 (Figure 2). This represents a reduction relative to the Joseph estimate. It also reflects hedge-based, no-arbitrage pricing rather than belief-based averaging.

The two pricing approaches lead to different option values, and that gap allows Mark to structure a favorable trade. Mark builds a position using shares and borrowed funds that produces the same payoff as the option whether the stock goes up or down. Mark buys 9,125 shares and finances the rest with a loan at 5 percent, then sells 10,500 options to Joseph at Joseph's estimated price of \$5.21 each, bringing in about \$54,705 in premium. He builds the position so that the combination of stock ownership and borrowing offsets what he might owe on the options. When the numbers are worked through, Mark ends up making about \$27,330 (Figure 3) whether the stock goes up or down. Since the result is the same in either case, his profit is essentially locked in and comes directly from the fact that the options were priced too high.

Joseph's side of the trade is much less certain. He pays about \$54,705 for the options and takes the risk that they could expire worthless. If the stock rises to \$40.95, the options pay about \$114,975 and he clears roughly \$60,270 after subtracting what he paid. If the stock drops, the options expire and he loses the full premium. Using his 50/50 assumption, the average expected profit is only about \$2,760 (Figure 3). Although the expected payoff is marginally positive under his assumptions, the full premium remains at risk if the stock declines.



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Marks / Joseph Option Deal Financial Summary	
Inputs	
Personal Investment	\$27,375.00
Annual interest rate	5.00%
Interest amount (given)	\$12,318.75
Option premium per share	\$5.21
Contract shares (units)	10,500
Strike price	\$30.00
Market price (up case)	\$40.95
Owned shares initially	9125
Down-case sale price	\$28.35
Loan Principal	\$246,375.00
Premium received by Mark (given)	\$54,705.00
Mark's Profit / Loss	
UP CASE	
Premium received	\$54,705.00
Proceeds from selling (Up stock price * owned shares)	\$373,668.75
Less cost of original shares	\$(273,750.00)
Less loss on options sold	\$(114,975.00)
Less interest on loan	\$(12,318.75)
Net proceeds (UP)	\$27,330.00
DOWN CASE	
Premium received	\$54,705.00
Proceeds from selling (Down price×Owned)	\$258,693.75
Less cost of original shares	\$(273,750.00)
Less loss on options sold	\$-
Less interest on loan	\$(12,318.75)
Net proceeds (DOWN)	\$27,330.00
Joseph's Option Math	
Personal Investment (premium paid)	\$5.21
Option premium per share (Joe)	\$5.21
Contract shares (units)	10,500
Strike price	\$30.00
Up stock price	\$40.95
Down stock price	\$28.35
Total premium paid	\$54,705.00
Option payoff if UP	\$114,975.00
Option payoff if DOWN	\$-



Profit if UP (Payoff–Premium)	\$60,270.00
Expected payoff ($p = 0.5$)	\$57,487.50
Expected net profit ($p = 0.5$)	\$2,782.50

Figure 3 Financial Summary of the options between Mark and Joseph

Overall, this comparison shows that the simple averaging approach pushes the option value too high. The binomial model produces a lower value because it is based on what it would actually cost to match the option payoff using stock and borrowing. Mark's ability to earn the same profit in every outcome is strong evidence that the lower, hedge-based price is the more reliable value, while Joseph's higher price gives the other side an easy advantage.

4. RECOMMENDATION

Since we do not have information regarding Joseph's other job offers, our recommendation for Joseph is based on the overall competitiveness of Joseph's compensation package and his tolerance risk. If Joseph believes the rest of his compensation package is reasonably competitive and he is confident in his expectations for MHP's annual upward and downward stock price movement over the next four years, he should take the offer and secure the 20,000 stock options. However, Joseph should not buy an additional MHP option from Mark as his financial prospects will already be heavily dependent on the performance of MHP. While there is certainly some risk that his company sponsored options expire worthless, he is not exposed to any direct downside risks as he does not have to purchase the options. On the other hand, if he were to take Mark's offer now not only is a presumably significant portion of his compensation contingent on MHP performing well over the next four years, but now also his personal portfolio is staked on one company's short-term performance. Options are generally not advisable for a retail investor, and in this scenario, we do not recommend naked options (unhedged) for Joseph.

For Joseph's employee stock options, our binomial asset pricing model tells a clear story. The expected value of his options in 4 years in today's dollars is \$138,000. In other words, Joseph can expect his total compensation over the next four years to also include ~\$34,500 per year in option gains. Joseph should expect significant volatility and should not over index on assessing his option generated compensation on an annual basis. Luckily for Joseph, his options do not vest for four years, so he will be forced to hold his options regardless of fluctuations that may tempt him to "take the money" if MHP's stock were to say gain 30% each year for the first two years. Either way, Joseph should treat any option generated compensation as a bonus and should not account for this money as he plans his personal finances and expense burden over the next four years. Our binomial asset model does not account for unforeseen macro conditions, for example, 911, the Global Financial Crisis, or Covid-19. External forces could at any moment erode the value of his options and render them worthless at the end of his four-year vesting period. All in all, despite the sizable expected value of Joseph's employee stock options, it is best to plan for no upside.

Unlike Joseph, Mark appears to be taking on leverage by borrowing against his share position to execute the hedge. The profitability of the strategy comes from Joseph's pricing assumptions. Joseph values the option at \$19.55 per share, while the binomial model indicates a value of \$6.90 per share. That \$12.65



difference per contract results in an overvaluation of roughly \$253,000 across 20,000 options. By selling the options at Joseph's higher price and simultaneously constructing the offsetting share position, Mark captures that spread. In summary, selling the options at Joseph's price represents an arbitrage opportunity for Mark. For Joseph, the trade offers limited expected benefits relative to the risk and should not be pursued at that price.

5. ACTION PLAN

Now that Joseph has piqued his interest in asset valuation modeling, he should continue studying no-arbitrage pricing, risk-neutral probability, and replicating portfolio construction. These concepts form the foundation of modern option valuation. Time value of money, present value, and future value formulas are taught in introductory finance courses and offer a strong starting point for building this foundation. Joseph could reference structured online resources such as Khan Academy to strengthen his understanding of these fundamentals before progressing into more advanced material. Once he strengthens these fundamentals, Joseph should build a multi-period binomial model in Excel and test how changes in volatility, interest rates, or stock movement assumptions impact the option value. Conducting sensitivity analysis will allow him to see how small changes in assumptions can materially affect pricing outcomes.

After developing comfort with the binomial framework, Joseph could compare his results to a Black-Scholes valuation to confirm consistency. Revisiting more advanced materials, including stochastic calculus for finance, will also provide better insight into why risk-neutral probability adjustments are necessary to eliminate arbitrage opportunities. In addition, Joseph should evaluate how concentrated exposure to MHP stock affects his overall financial position, particularly given vesting restrictions and the possibility of employment changes. By combining structured study, formal instruction, and practical modeling, Joseph will be better equipped to evaluate both this opportunity and future equity-based compensation decisions.

6. APPENDIX

A. Model Parameters			
Current Stock Price S0	\$ 30.00	USD	
Risk-Free Rate r	0.05	per year	
Up Factor u = 1.05*1.30	1.365		
Down Factor d = 1.05*0.90	0.945		
Number of Years n	4	years	
Strike Price K	\$ 30.00	USD	
B. Year-4 Stock Price Lattice and Terminal Payoffs			
k (up moves)	S4(k)	Call Payoff C4(k)=max(S4-K,0)	
0	23.925	0	
1	34.558	4.558	
2	49.917	19.917	
3	72.103	42.103	
4	104.148	74.148	
C. Joseph's Expected Values (50/50 per year)			
E(S4)	\$ 53.39	USD	
PV(E(S4)) @5%	\$ 43.92	USD	
E(C4)	\$ 23.77	USD	
PV(E(C4)) @5%	\$ 19.55	USD	
D. Risk-Neutral Option Valuation (Binomial)			
Risk-neutral probability p	0.25		
Discount factor DF = 1/(1+r)	0.95		
No-arbitrage option price C0	\$ 6.90	USD	
E. Outputs			
Expected Stock Value in Year 4	\$ 53.39	USD	
Present Value of Expected Stock	\$ 43.92	USD	
Expected Option Payoff in Year 4	\$ 23.77	USD	
Present Value of Expected Option Payoff	\$ 19.55	USD	
Risk-Neutral (Fair) Option Price	\$ 6.90	USD	

Figure 1 Examples of calculations